

Impact of Physician Integration

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Introduction

Over the past several decades, the environment in which physicians deliver patient care has transformed, with more physicians seeking employment by larger healthcare organizations rather than entering or remaining in private practice. The American Medical Association (AMA) Physician Practice Benchmark Survey, a nationally representative survey of physicians conducted biennially, found that in 2024 only 42.2% of physicians worked in private practices, an 18-percentage-point decline from 2012. During the same period, the share of physicians working in hospital-owned practices increased from 23.4% to 34.5%, while 12.2% of physicians were directly employed by hospitals in 2024, more than double the 5.6% reported in 2012. Private-equity ownership has also increased, with 6.5% of physicians reporting that their practices were owned by private-equity firms in 2024. These trends reflect a substantial change in the organizational structure of physician practice (Kane, 2025, p. 10, Exhibit 1).

Several factors may help explain this shift. Physicians joining larger organizations may gain access to costly resources and capital, centralized administrative support, and greater negotiating leverage with payers. Integration may therefore generate economies of scale and reduce some of the operational burdens associated with independent practice. At the same time, increased provider concentration may strengthen bargaining power and contribute to higher physician prices. These competing effects—economies of scale versus market power—form the economic framework for this analysis. The central question is whether measurable efficiencies from physician integration are sufficiently large to offset the effects of increased market power. Physician prices serve as the primary empirical price measure, while site of care, spending, operational efficiency, and patient outcomes provide supporting evidence of the broader effects of integration.

Economic Framework and Price Evidence

Economic theory does not necessarily predict whether physician integration will increase or decrease healthcare costs. The net effect depends on whether efficiencies from integration outweigh the effects of increased market power. Gaynor and Haas-Wilson provide this industrial-organization framework: efficiencies from consolidation are possible but are not automatic, and the net price effect depends on the relative strength of efficiency and market-power effects.

Cooper et al. provide the primary empirical benchmark for evaluating the price effect. Examining hospital acquisitions of physician practices between 2008 and 2016, they found that two years after integration, physician prices for childbirth admissions increased 15.1% (approximately \$502) and hospital labor-and-delivery prices increased 3.3% (\$475), without a discernible improvement in the quality measures studied. Importantly, physician prices also increased approximately 9% among physicians who were already integrated when their hospital acquired additional physicians in the same specialty. Because these physicians did not themselves change integration status, the finding supports increased physician-market concentration and bargaining leverage as an explanation for at least part of the observed price effect (see Figure 1).

Referral and Site-of-Care Mechanisms

The price and spending effects may also operate through changes in referral patterns and site of care. Cooper et al. identify referral redirection to acquiring hospitals, increased hospital bargaining leverage through a “recapture” mechanism, and horizontal physician-market concentration as mechanisms associated with integration. Whaley and Zhao provide complementary evidence by examining referral patterns following physician integration. After integration, PCPs were noted to have a sharp and immediate increase in the use of HOPDs (hospital outpatient department) of approximately 5 percentage points. They also found that HOPD use increased by 5.0 percentage points for arthroscopy and 6.8 percentage points for colonoscopy following integration. These changes translated into relative increases of 8.2% and 9.3% respectively. These shifts moved procedures toward higher-reimbursement hospital outpatient settings and increased Medicare and patient spending without measurable improvement in complication rates for the procedures studied.

Together, these studies illustrate two related pathways through which integration may affect healthcare costs. Greater physician concentration can strengthen bargaining leverage and increase negotiated physician prices, while ownership and referral incentives can shift services toward higher-cost sites within the integrated system. Referral steering itself does not establish that a referral is clinically inappropriate;

rather, the economic concern is whether ownership changes referral incentives in ways that increase spending without a corresponding improvement in patient outcomes.

Do Efficiencies Offset the Price Effects?

Evidence of higher prices or spending must be weighed against any measurable efficiencies or improvements in outcomes associated with integration. Saghaian et al. provide an important example of this tradeoff. Using national patient-level data covering approximately 2.6 million Medicare visits across 5,488 physicians, the study found increased physician throughput following gastroenterology integration, indicating improved operational efficiency. At the same time, integration was associated with approximately \$127 greater Medicare spending per colonoscopy, largely driven by facility fees, and increased annual colonoscopy spending. The study also reported increases in several post-procedure complications. Thus, higher physician throughput represented an operational efficiency for the practice, not a demonstrated payer saving: Medicare spending increased by approximately \$127 per colonoscopy.

This pattern is consistent with Harris et al.'s systematic review of 43 hospital-physician integration studies. Among studies evaluating costs, most found hospital-physician integration to be associated with higher healthcare costs. Factors contributing to higher costs included higher physician prices in commercial markets and shifts toward more expensive care settings in Medicare markets. In contrast, evidence of improved utilization or quality was less consistent. Positive quality findings were largely associated with process measures, particularly in cardiovascular care, while other studies found no improvement or negative effects on quality. Overall, the review found higher costs to be considerably more consistent than measurable improvements in quality or utilization.

Taken together, the evidence indicates that physician integration can produce measurable operational efficiencies; however, the studies reviewed do not demonstrate that these efficiencies consistently translate into lower healthcare spending or improved patient outcomes. These findings provide the basis for evaluating whether the documented efficiencies are economically large enough to offset the observed increases in prices and spending.

Net Economic Assessment and Policy Implications

Taken together, the evidence reviewed does not meet the predetermined conditions that would challenge the working hypothesis. First, the price condition is not met. Rather than demonstrating prices at or below those of comparable independent practices, Cooper et al. found physician prices for childbirth admissions increased 15.1% (approximately \$502) two years after integration.

Second, although physician integration can generate operational efficiencies, the efficiency-offset condition is not met. The studies reviewed did not demonstrate savings per patient or episode of approximately the same economic magnitude as the additional spending associated with integration. Harris et al.'s systematic review supports this assessment, finding higher costs more consistently than measurable efficiency or quality gains.

The quality-offset condition also is not met. The evidence reviewed did not demonstrate sufficiently consistent and clinically meaningful improvements in patient outcomes to compensate for the observed increases in prices and spending. This does not mean that physician integration produces no efficiencies or quality benefits. Rather, based on the evidence reviewed, those benefits were not demonstrated to be sufficiently large or consistent to offset the observed price and spending effects. The evidence therefore supports the working hypothesis: physician integration may generate genuine efficiencies, but in the studies reviewed, the effects associated with increased market power appear to outweigh the measurable economic offsets.

These findings have implications for antitrust review. The Federal Trade Commission is aware that mergers can raise antitrust concerns resulting from reduced competition and rising healthcare costs. As an example of broader federal and state antitrust oversight of healthcare consolidation, the FTC and Rhode Island Attorney General challenged the prospective merger of Lifespan and the Care New England Health system, which would have combined Rhode Island's two largest healthcare providers into one entity controlling at least 70% of inpatient patient care. Consequently, both parties abandoned the merger. Unlike a large hospital merger that is readily identifiable for antitrust review, individual physician-practice acquisitions may be relatively small, yet repeated acquisitions within a specialty or local market can cumulatively increase concentration and bargaining power. The FTC and state antitrust regulators should explicitly consider changes in physician-market concentration when evaluating hospital acquisitions of physician practices, including acquisitions that may be individually small but cumulatively increase

concentration within a local market. FTC resources and staffing should be augmented to be able to perform these reviews. State oversight also warrants attention because mechanisms such as Certificates of Public Advantage (COPAs) can provide state-action immunity from federal antitrust law and immunity from private claims under state antitrust laws, potentially limiting federal review of otherwise anticompetitive consolidation. Cooper et al.'s finding that prices increased even among already-integrated physicians when hospitals acquired additional physicians in the same specialty suggests that competitive effects may extend beyond the physicians directly involved in an acquisition. Greater attention to cumulative physician-market concentration would allow regulators to evaluate whether expected efficiencies from integration are likely to outweigh potential effects on bargaining power and prices.

Conclusion and Limitations

Physician integration may produce measurable operational efficiencies but the literature that this analysis is based on indicates that efficiencies may not offset the higher healthcare costs that are created after integration. This analysis found that higher physician prices were found after integration, but physician prices themselves are not synonymous with total healthcare spending and that integration effects can vary across markets and specialties. It is therefore recommended that efficiencies should be demonstrated rather than assumed when evaluating provider integration systems. Finally, prospective mergers and integrations should undergo more extensive federal and state oversight when cumulative consolidation increases market concentration and bargaining power without demonstrated efficiencies sufficient to offset higher prices or spending.

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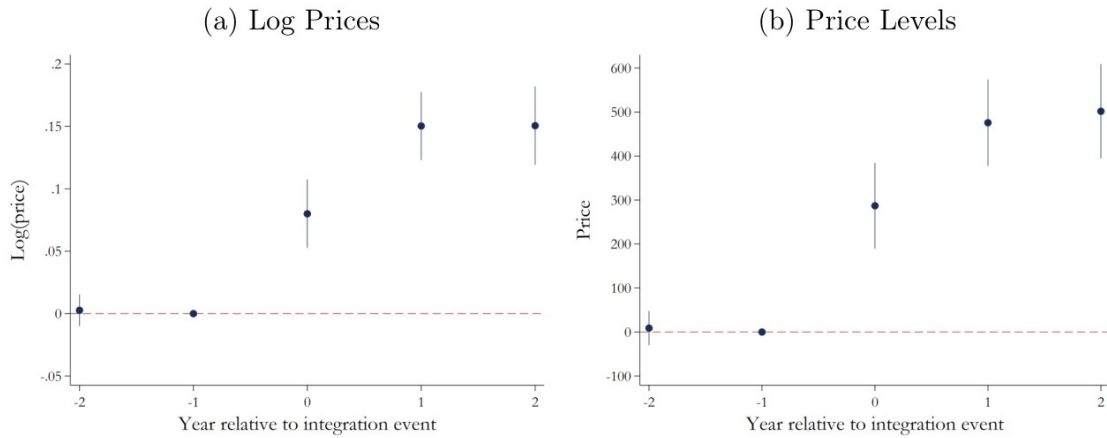
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Figure 1

Physician Prices Around Physician Integration

those observed for the hospital-level events, in part due to the hospital event sample including many non-integrating physicians.

Figure 6: Event Study: Physician Prices Around Physician Integration Event



Notes: Physician-level event studies comparing prices of treated physicians to those of matched controls, in the five-year period covering the event. The dependent variables are log price in panel (a) and price in panel (b).

Note. Reproduced from Cooper et al. (2025), Figure 6, Are Hospital Acquisitions of Physician Practices Anticompetitive? NBER Working Paper No. 34039. This is the principal empirical price figure used in the analysis.